

Certified analytical bounds for hypersonic reachable sets

A two-page summary of a preliminary PhD research proposal

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Full proposal: research.ekdynamics.science/proposal.pdf

The result. This is a problem in nonlinear analysis: a singularly perturbed family of differential inclusions in which ordinary time-parametrised limits do not retain the concentrated state transitions. The motion carrying them occupies a vanishing time, so it disappears from those limits while still deciding where the trajectories go. Compactness itself does not fail — the corridor is compact and the occupation measures stay weak-* compact. What fails is that a time-parametrised limit weighs an interval of time by its length. For a model keeping exactly that phenomenon I can prove the outer inclusion below — an estimate a profile decomposition would have to reproduce, and not itself one:

$$\mathcal{R}_\varepsilon(T) \subseteq \left[\underbrace{\mathcal{R}_\mathcal{H}([T - \varrho, T + \varrho])}_{\text{skeleton, bubbles and glide}} + \Lambda \times \{0\} \right] \oplus \underbrace{\mathbb{B}(\varrho)}_{\text{residual}},$$

where Λ lifts a state of the averaged mode to the fast fibre it stands for, and the residual is a sum of three terms — the drift dropped over the passes, the error of averaging the fast part, and the speed error the passes route into the slow motion — each bounded in closed form, with every constant explicit in the data and free of ε . A single amplitude threshold sets the balance between the three, and at its best choice $\varrho = O((\varepsilon \log(1/\varepsilon))^{1/5})$ — I claim the order and not its sharpness — for every ε below a threshold explicit in the data. What the rate turns on is the *concentration* of the control's variation, not its size: a bounded slew is paid for by the vanishing duration of a pass; a switch falling inside one injects the same amount however brief the pass, and leaves a floor proportional to that switch. The inclusion is uniform over the controls and the initial set, and holds for every member of a bounded family of admissible fields. The proof is elementary estimates and exact integral identities; the statement, the hypotheses and the proof are in the proposal.

The problem. The structural phenomenon is concentration, and its natural language — not a prerequisite, as the theorem above shows — is a profile decomposition: a *skeleton*, one *bubble* per pass, and a *residual*. In the theory of nonlinear dispersive equations such a decomposition comes from iterating the dichotomy of the concentration-compactness principle [1, 2], and what it delivers is structure without a constant — by contradiction. The proposal is that here the structure can be had quantitatively, because the family carries Lyapunov functionals: the dissipation is monotone with total variation bounded uniformly in the parameter, so the number of profiles above any fixed size is bounded in advance rather than extracted. That count is proved; the decomposition it would organise is not. What I think is interesting is that the concentration structure turns out to be reachable without the decomposition machinery at all, by an exact integral identity and a budget. Whether the machinery buys anything beyond that is the first question of the programme.

The example that supplies the problem is hypersonic atmospheric entry, where the solution set is the footprint a vehicle can reach and the parameter is the ratio of the atmospheric scale height to the planetary radius. It is the example and the test, not the subject. Computing the set exactly means a Hamilton–Jacobi–Isaacs equation on a grid, exponential in the state dimension and capped in practice near four or five states [3], while a rigid-body entry vehicle carries thirteen.

The model. The state is (z, v, s) , with z an altitude, v its rate and s a vector of slow coordinates, and

$$\dot{z} = v, \quad \dot{v} = c m^\varepsilon(z) - 1 - \kappa v, \quad \dot{s} = F + m^\varepsilon(z) \Gamma, \quad m^\varepsilon(z) = m(z/\varepsilon),$$

with m positive and strictly decreasing and c a scalar control: a unit-mass particle in the potential Φ_c with $\Phi'_c = 1 - c m^\varepsilon$, one soft wall and uniform gravity above it. The small parameter sits in one place only, the width of the band in which the air is felt; the equations of motion carry none. All of m, κ, F, Γ may be chosen from a bounded family — one choice per flight, so the density is enclosed rather than fixed while the control still ranges over a set at each instant. Letting it vary *during* a flight enlarges the enclosure and is an open problem below: the potential depends on that choice, so a time-varying one moves the well beneath the oscillation and the amplitude stops obeying a closed law. The budget, a common Lyapunov functional, is indifferent to this; the amplitude is not.

The reduced oscillator retains the altitude-rate dynamics, the atmospheric forcing, the dissipative coupling and the slow-state forcing carrying the singular mechanism, and three regimes of flight are three energies $W = \frac{1}{2}v^2 + \Phi_c(z)$ in one well: the minimum is the equilibrium glide, a small oscillation is the phugoid, a large one is a skip. *The pass is short and the arc is long because the well is asymmetric*, and that asymmetry, read at the scale of the band, is the whole of the concentration. Two Lyapunov functionals do the quantitative work — the amplitude W for the fast motion and the dissipation for the slow — and three exact identities organise it. First, ε sets no shape, only a clock, read at amplitude W/ε — which is why a bounded oscillation keeps a period of order one while the part of it in the

band shrinks like $\varepsilon \log(1/\varepsilon)$. Second, integrating the rate equation across a pass gives the impulse it delivers as twice its entry speed over c , exactly — so a pass takes a vanishing share of the time and essentially all of the effect, and the budget admits only so many such passes. Third, the same rate equation gives the oscillating part of the density as a derivative, which is what bounds the averaging error: where a periodic argument would invoke a zero mean over a closed orbit, here the primitive is controlled by the velocity excursion, so no orbit need close and neither the control nor the damping need be frozen. The trajectory is cut at its own crossings of $z = 0$ and nowhere else. The limit system $\mathcal{H}$ flows ballistically on the arcs, jumps at $z = 0$ by a map J — the solution set of an inner problem whose slow equation, written in the delivered impulse, carries no small parameter — and flows by the averaged field $F + \Gamma/c$ elsewhere. The hypotheses, the constants and the proof are in the proposal.

What makes the *estimates* elementary is that the functionals do all the counting: the passes above a fixed size, their summed duration, and the accumulated amplification, which is exponential in each pass’s own impulse and so bounded however many there are. One discipline runs throughout: the density is unbounded inside the band as $\varepsilon \rightarrow 0$ and is never bounded pointwise there, only by its impulse — exactly what the dissipation functional controls.

Where the measures enter. The theorem hands the occupation-measure programme of Lasserre, Henrion and collaborators [4] a different problem: $\mathcal{R}_{\mathcal{H}}$ is a reachability problem in $2 + d$ coordinates with a set-valued reset and a free mode schedule, near the hybrid form whose relaxation is known [5]. The Liouville equation constrains a measure linearly, a measure on the reset carries the mass across each pass, and the dual certificate is a pair of functions non-decreasing along the flow and across it. Whether the limit is better taken in that topology or at the level of trajectories is open.

What remains. The theorem applies to entry, but its residual is not small at the physical ε . A pass entered at flight-path angle γ_e enters through the ratio $\gamma_e/\sqrt{\varepsilon}$, as in the classical Chapman variables [6], and for the Earth $\varepsilon \approx 1.3 \times 10^{-3}$, so $\sqrt{\varepsilon} \approx 0.036$ radians or about two degrees, while entry angles are a few degrees. So the amplitude at which a pass becomes brief is the amplitude at which the excursion being averaged becomes large, and no threshold makes both small. Four problems follow.

1. **The decomposition itself.** That every sequence of trajectories splits, along a subsequence, into the skeleton, finitely many profiles obtained by rescaling the passes to their own widths, and a remainder small in a stated norm. The count such a decomposition needs is proved; the decomposition is not.
2. **The statement at a fixed ε .** The motion near the threshold, which neither approximation serves, must be carried by the unapproximated field. That adds no term to the residual, since nothing is approximated; it costs dimension, over a duration the budget bounds. What has to be shown is a bound on the growth of that enclosure, explicit in the interval widths — and with it the class of time-varying atmospheres for which the amplitude law survives, bounded variation in altitude being what the physics supplies.
3. **Separation of scales, by enclosure.** The attitude and elastic motions do not concentrate, and averaging them is the obvious route and the wrong one: an averaging theorem gives an order and not a bound, and holds through a resonance only outside a set of small measure, which an outer bound cannot discard. They act on the trajectory only through the aerodynamic force, so the force is what must be enclosed. Two facts are to be established. Where there is no air there is neither torque nor aerodynamic force, so the attitude cannot act on the trajectory. Where there is air, the damping rate goes as density times speed, while the energy identity makes the integral of density against the *cube* of the speed exactly the energy shed; dividing by the square of the largest admissible speed turns that equality into a lower bound on the integrated damping.
4. **The companion at the level of measures.** That the limit of the occupation measures satisfies the Liouville equation of the limit system, with the mass crossing each pass carried on the graph of J . This is separate from the first problem, and is where the concentration machinery is expected to be needed — for the structure, not the constants.

The first and fourth are the analysis; the second and third carry the model to the application. The fourth is what I would most value a conversation about: granting the limit of the measures and the profiles it carries, whether the limiting solution set is contained in the one those profiles generate is a further implication, and the hardest link in the chain — the part for which I am looking for an advisor.

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| <p>[1] P.-L. Lions. “The Concentration-Compactness Principle in the Calculus of Variations. The Locally Compact Case, Part 1”. In: <i>Annales de l’Institut Henri Poincaré, Analyse Non Linéaire</i> 1.2 (1984).</p> <p>[2] H. Bahouri and P. Gérard. “High Frequency Approximation of Solutions to Critical Nonlinear Wave Equations”. In: <i>American Journal of Mathematics</i> 121.1 (1999).</p> <p>[3] S. Bansal et al. “Hamilton-Jacobi Reachability: A Brief Overview and Recent Advances”. In: <i>56th IEEE Conference on Decision and Control (CDC)</i>. 2017.</p> | <p>[4] J.-B. Lasserre et al. “Nonlinear Optimal Control via Occupation Measures and LMI Relaxations”. In: <i>SIAM Journal on Control and Optimization</i> 47.4 (2008).</p> <p>[5] V. Shia et al. “Convex Computation of the Reachable Set for Controlled Polynomial Hybrid Systems”. In: <i>53rd IEEE Conference on Decision and Control (CDC)</i>. 2014.</p> <p>[6] D. R. Chapman. <i>An Approximate Analytical Method for Studying Entry into Planetary Atmospheres</i>. Tech. rep. Technical Note 4276. National Advisory Committee for Aeronautics, 1958.</p> |
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